Decision Theory

The decisions are classified according to the *degree of certainty* as **deterministic models**, where the manager *assumes* complete certainty and each strategy results in a **unique payoff,** and **Probabilistic models**, where each strategy leads to *more than one payoffs* and the manager attaches a *probability measure* to these payoffs. The scale of assumed certainty can range from complete certainty to complete uncertainty hence one can think of **decision making under certainty (DMUC) and decision making under uncertainty (DMUU)** on the two extreme points on a scale. The region that falls between these extreme points corresponds to the concept of probabilistic models, and referred as **decision-making under risk (DMUR)**. Hence we can say that most of the decision making problems fall in the category of decision making under risk and the assumed degree of certainty is only one aspect of a decision problem. The other way of classifying is: Linear or non-linear behaviour, static or dynamic conditions, single or multiple objectives. One has to consider all these aspects before building a model.

Decision theory deals with decision making under conditions of risk and uncertainty. For our purpose, we shall consider all types of decision models including deterministic models to be under the domain of decision theory. In management literature, we have several quantitative decision models that help managers identify optima or best courses of action.

Complete uncertainty	*Degree of uncertainty*	*Complete certainty*
Decision making	Decision making	Decision-making
Under uncertainty	**Under risk**	**Under certainty.**

Before we go to decision theory, let us just discuss the issues, such as (*i*) What is a decision? (*ii*) Why must decisions be made? (*iii*) What is involved in the process of decision-making? (*iv*) What are some of the ways of classifying decisions? This will help us to have clear concept of decision models.

WHAT IS A DECISION?

A **decision** is the conclusion of a process designed to weigh the relative utilities or merits of a set of available alternatives so that the most preferred course of action can be selected for implementation. Decision-making involves all that is necessary to identify the most preferred choice to satisfy the desired goal or objective. Hence decision-making process must involve a set of goals or objectives, a system of priorities, methods of enumerating the alternative courses of feasible and viable courses and a system of identifying the most favourable alternative. One must remember that the decisions are sequential in nature. It means to say that once we select an alternative, immediately another question arises. For example if you take a decision to purchase a particular material, the next question is how much. The next question is at what price. The next question is from whom… Like that there is no end.

WHY MUST DECISIONS BE MADE?

In management theory we study that the essence of management is to make decisions that commit resources in the pursuit of organizational objectives. Resources are limited and wants and needs of human beings are unlimited and diversified and each wants to satisfy his needs in an atmosphere, where resources are limited. Here the decision theory helps to take a certain decision to have most satisfactory way of satisfying their needs. Decisions are made to achieve these goals and objectives.

DECISION AND CONFLICT

When a group of people is working together in an organization, due to individual behaviour and mentality, there exists a conflict between two individuals. Not only that in an organization, each department has its own objective, which is subordinate to organizational goal, and in fulfilling departmental goals, there exists a conflict between the departments. Hence, any decision maker has to take all these factors into consideration, while dealing with a decision process, so that the effect of conflicts between departments or between subordinate goals is kept at minimum in the interest of achieving the overall objective of the organization.

TWO PHASES OF THE PROCESS OF DECISION-MAKING

The decision theory has assumed an important position, because of contribution of such diverse disciplines as philosophy, economics, psychology, sociology, statistics, political science and operations research to the area decision theory. In decision-making process we recognize two phases: (1) How to formulate goals and objectives, enumerate environmental constraints, identify alternative strategies and project relevant payoffs. (2) Concentration on the question of how to choose the optimal strategy when we are given a set of objectives, strategies, payoffs. We concentrate more on the second aspect in our discussion.

CLASSIFICATIONS OF DECISIONS

In general, decisions are classified as **Strategic decision**, which is related to the organization's outside environment, **administrative decisions** dealing with structuring resources and operational decisions dealing with day-to-day problems. Depending on the nature of the problem there are **Programmed decisions**, to solve repetitive and well-structured problems, and **Non-programmed decisions**, designed to solve non-routine, novel, ill structured problems. Depending on the scope, complexity and the number of people employed decision can be divided as **individual and managerial** decisions. Depending on the sphere of interest, as **political**, **economic**, or **scientific** etc. decision can be divided as **static** decision requiring only one decision for the planning horizon and **dynamic decision** requiring a series of decisions for the planning horizon.

STEPS IN DECISION THEORY APPROACH

1. List the **viable alternatives (strategies)** that can be considered in the decision.

2. List all **future events** that can occur. These future events (not in the control of decision maker) are called as **states of nature**.
3. Construct a payoff table for each possible combination of alternative course of action and state of nature.
4. Choose the criterion that result in the largest payoff.

DECISION MAKING UNDER CERTAINTY (DMUC)

Decision making under certainty assumes that all relevant information required to make decision is certain in nature and is well known. It uses a deterministic model, with complete knowledge, stability and no ambiguity. To make decision, the manager will have to be quite aware of the strategies available and their payoffs and each strategy will have unique payoff resulting in certainty. The decision-making may be of single objective or of multiple objectives.

Problem – 1:

ABC Corporation wants to launch one of its mega campaigns to promote a special product. The promotion budgets not yet finalized, but they know that some Rs. 55,00,000 is available for advertising and promotion.

Management wants to know how much they should spend for television spots, which is the most appropriate medium for their product. They have created five 'T.V. campaign strategies' with their projected outcome in terms of increase in sales. Find which one they have to select to yield maximum utility. The data required is given below.

Strategy	*Cost in lakhs of Rs.*	*Increased in sales in lakhs of Rs.*
A	1.80	1.78
B	2.00	2.02
C	2.25	2.42
D	2.75	2.68
E	3.20	3.24

Solution

The criteria for selecting the strategy (for maximum utility) is to select the strategy that yields for maximum utility *i.e*. highest ratio of outcome *i.e*. increase in sales to cost.

Strategy	*Cost in Lakhs of Rs.*	*Increase in Sales in Lakhs of Rs.*	*Utility or Payoffs*	*Remarks.*
A	1.80	1.78	1.78 / 1.80 = 0.988	
B	2.00	2.02	2.02 / 2.00 = 1.010	
C	2.25	2.42	2.42 / 2.25 = 1.075	**Maximum Utility**
D	2.75	2.68	2.68 / 2.75 = 0.974	
E	3.20	3.24	3.24 / 3.20 = 1.012	

The company will select the third strategy, *C*, which yields highest utility.
Now let us consider the problem of making decision with multiple objectives.

Problem 2:

Consider a M/s *XYZ* company, which is developing its annual plans in terms of three objectives: (1) Increased profits, (2) Increased market share and (3) increased sales. M/S *XYZ* has formulated three different strategies for achieving the stated objectives. The table below gives relative weightage of objectives and scores project the strategy. Find the optimal strategy that yields maximum weighted or composite utility.

Measure of → *Performance of* *Three objectives*	*ROI* *(Profit)*	*% Increase* *(Market share)*	*% Increase* *(Sales growth)*
Weights →	0.2	0.5	0.3
Strategy			
S_1	7	4	9
S_2	3	6	7
S_3	5	5	10

Solution

(The profit objective could be stated in and measured by absolute Rupee volume, or percentage increase, or return on investment (ROI). The market share is to be measured in terms of percentage of total market, while sales growth could be measured either in Rupees or in percentage terms. Now, in order to formulate the payoff matrix of this problem, we need two things. (*i*) We must assign relative weights to each of the three objectives. (*ii*) For each strategy we will have to project **a score** in each of the three dimensions, one for each objective and express these scores in terms of utilities. The Optimal strategy is the one that yields the maximum weighted or composite utility.)

Multiplying the utilities under each objective by their respective weights and then summing the products calculate the weighted composite utility for a given strategy. For example:

For strategy $S_1 = 7 \times 0.2 + 5 \times 0.5 + 9 \times 0.3 = 6.1$

Measure of → *Performance of* *Three objectives*	*ROI* *(Profit)*	*% Increase* *(Market share)*	*% Increase* *(Sales growth)*	*Weighted or* *Composite* *Utility (CU)*
Weights →	0.2	0.5	0.3	
Strategy				
S_1	7	4	9	$0.2 \times 7 + 0.5 \times 4 + 0.3 \times 9 = 6.1$
S_2	3	6	7	$0.2 \times 3 + 0.5 \times 6 + 0.3 \times 7 = 5.7$
S_3	5	5	10	$0.2 \times 5 + 0.5 \times 5 + 0.3 \times 10 = 6.6$ Maximum utility